

A bioactivity graph that knows what binding *does*

~~shrine-diet-bioactivity~~ → shrine-bioactivity

The graph as built is an excellent *occurrence* and *potency* substrate: what compound is in what food, how tightly it binds, how often the pair is cited. It has almost no representation of *mechanism*. This document maps what exists, verified against source, and proposes the amendment that turns it into a mechanism-of-action knowledge base with a semantic-context layer as the driver.

Repo Syntropy-Health/shrine-diet-bioactivity Commit db3c740

Schema source graphvdb/entity_schema.py (948 L)

Revision v2 – incorporates design review (§14)

GraphVDB is the role; the vendor is an implementation detail

Throughout this document **GraphVDB** names the graph-plus-vector retrieval substrate: the component that holds typed entities and edges alongside their embeddings and answers hybrid structural + semantic queries. It is deliberately a *role name*, not a product name.

The substrate is supplied by a third-party framework, subclassed by the scoped wrapper. Which framework is a build-time binding, not an architectural commitment, so it is left unnamed here: nothing in this design depends on it and naming it would smuggle a vendor back into the spec. On disk the modules implementing it still sit under `lightrag/`; every `graphvdb/...` path in this document resolves there today.

A9 proposes renaming the path to match the role, alongside the repo rename — the vector tier has *already* been abstracted this way via the `KG_VECTOR_BACKEND` switch, so the graph tier is finishing a move that is half done.

01 The rename is a scope decision, not a string replace

Dropping `diet` from the name widens the mission axis. Today the provenance doc states it explicitly: "*every node and edge serves the diet $\rightleftharpoons$ food $\rightleftharpoons$ bioactive compound $\rightleftharpoons$ symptom $\rightleftharpoons$ disease retrieval objective; the KG is intentionally narrow on this axis.*"

Several real datasets were **deliberately refused** under that narrowness — the bulk TTD drug-disease table (30K rows) was rejected as "off-mission per parsimony: drugs aren't diet."

Generalizing to `shrine-bioactivity` reverses that judgment. Diet becomes one *exposure route* into a compound-centric bioactivity graph, alongside supplement, botanical extract, and drug. That is the right move for an MoA knowledge base — MoA is defined on the compound, not on the plate it arrived on — but the parsimony principle that governed six prior ingest decisions has to be formally superseded, not quietly ignored.

RECOMMENDED

Write **ADR 0011 — mission axis generalization** superseding the parsimony clause in `2026-04-29-mcp-gateway-design.md §8.3`. Without it, the next ingest argument re-litigates from scratch and the deferred-sources table stays frozen at a judgment the rename has invalidated.

RENAME DEBT, MEASURED

This repo has been renamed before and the last one was never finished — `shrine-open-diet` still appears in tracked files, including the live secret-rotation command in `CLAUDE.md`. Counts are tracked files containing each string at `db3c740`:

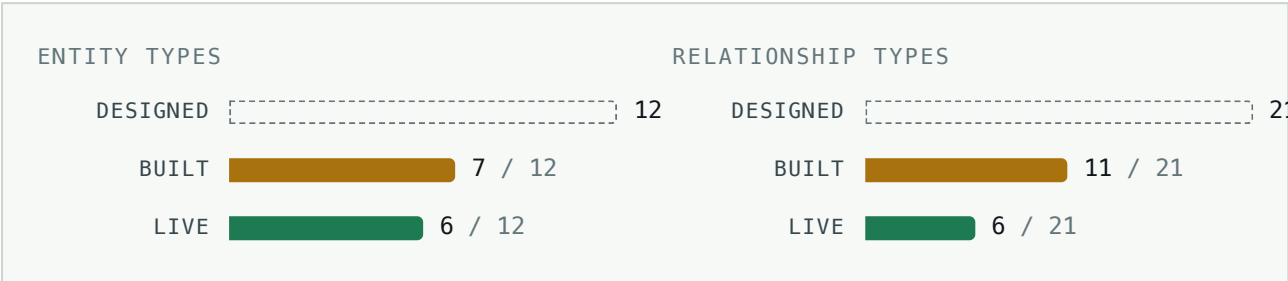
NAME-MIGRATION SURFACE			
STRING	FILES	WHAT IT IS	DISPOSITION
<code>shrine-diet-bioactivity</code>	110	Current repo + nested package + submodule path	Rename target
<code>herbal_botanicals</code>	59	SQLite filename <code>herbal_botanicals.db</code> , adapters, tests	Misnames the artifact — it holds foods, targets, diseases, pathways
<code>mcp-herbal-botanicals</code>	30	Pre-unification MCP name in legacy PRPs	Archaeology — leave, or sweep with the rename
<code>shrine-open-diet</code>	3	Prior repo name; one is a live <code>gh secret set --repo</code> command	Stale — will fail if run
<code>lightrag/</code>	2 paths	Vendored upstream submodule <i>and</i> our own module dir, same name	Vendor name in a structural path — see A9

Beyond files, the rename touches surfaces outside git: the monorepo submodule gitlink, the Railway app, the Infisical path `/research/shrine-diet-bioactivity`, mirrored GitHub Actions secrets, and the deployed hostname `kg-mcp-test.up.railway.app`. GitHub redirects the old repo URL, so the rename itself is low-risk; the risk is the *partial* rename, which is what the `shrine-open-diet` row above demonstrates already happened once.

Open issue [#8](#) already proposes renaming the nested Python package (52 files). Landing both in one sweep is cheaper than two migrations, and it is the moment to retire `herbal_botanicals.db` for a name that describes the 37 tables it actually holds.

02 Three tiers, and they do not agree

"The schema" is three different things depending on where you look, and the gaps between them are the most actionable facts in this document. **Designed** is what `entity_schema.py` declares. **Built** is what has a populated SQLite source table. **Live** is what the last graph snapshot actually found in Neo4j Aura.



DRIFT — UNVERIFIED FOR 3 MONTHS

The deployed graph predates Phases 2 through 5 entirely

The only graph-state snapshot in the repo (`research-journal/shared/scope-state-snapshot.md`) was generated **2026-04-29** and records exactly 7 relationship types — 6 real plus a legacy `DIRECTED` artifact. Every edge type built afterwards (`MAPS_TO_DISEASE` , the three evidence-typed compound→disease edges, `PATHWAY_INCLUDES_TARGET`) landed in SQLite on 2026-05-07/08. **Nothing in the repo records a re-ingest since.** The 2.92M-row evidence layer, the 24,403 canonical diseases, and the 370 KEGG pathways may exist only in SQLite.

This is stated as unverified rather than confirmed: I could not reach the live graph — the Aura and gateway credentials sit in an Infisical project this identity is not a member of. Re-running `graphvdb/generate_snapshot.py` resolves it in minutes and should precede every other decision in this document, because it determines whether the amendments below are additive or whether a full re-ingest is the prerequisite.

Drift runs in both directions. The live graph carries two node labels the schema never declares — Drug (35 nodes, from the curated herb-drug panel) and Gene (~21,000, from SymMap). `INTERACTS_WITH` declares `tgt_type: "Drug"` and `CONTRAINDICATES` declares `Condition|PatientState` , none of which exist in `ENTITY_TYPES` . The schema and the graph each contain things the other does not know about.

03 Entity schema — 12 declared

ENTITY_TYPES · GRAPHVDB/ENTITY_SCHEMA.PY				
ENTITY	SQLITE SOURCE	KEY / NAME FIELD	STATE	NOTE
CORE BOTANICAL-DIETARY BACKBONE				
Herb	herbs	scientific_name	● LIVE	2,376 Duke; ~9,000 Aura nodes across sources
Compound	compounds	name	● LIVE	94,512 rows; 0 SMILES, 0 populated PubChem CID
Food	compound_foods	food_name	● LIVE	962 nodes; 647 carry nutrition_100g JSON
Target	targets	name	● LIVE	4,355; UniProt + gene symbol + druggability
Disease	diseases_canonical	preferred_name	● BUILT	24,403 canonical; source swapped in Phase 3 — Aura still on the old multi-table path
Symptom	symptoms	name	● LIVE	47 hand-curated; SymMap holds 1,148 + 2,285 unjoined
EVIDENCE + MECHANISM OVERLAYS				
BioactivityEvidence	bioactivity_evidence	id	○ DORMANT	ChEMBL rows; Phase 1 ingest never run end-to-end
Pathway	kegg_pathways	name	● BUILT	370 human pathways; KEGG academic-licence only
CLINICAL / TENANT LAYER – DECLARED, NO SOURCE TABLE				
Protocol	none	name	⌂ DESIGNED	Tenant API ingest, never built
Intervention	none	name	⌂ DESIGNED	
Outcome	none	name	⌂ DESIGNED	
Biomarker	none	name	⌂ DESIGNED	
UNDECLARED – PRESENT IN THE LIVE GRAPH, ABSENT FROM THE SCHEMA				
Drug	hdi_safe_50.json	—	● LIVE	35 nodes; referenced by INTERACTS_WITH but never declared

ENTITY	SQLITE SOURCE	KEY / NAME FIELD	STATE	NOTE
Gene	SymMap	—	● LIVE	~21,000 nodes; the graph's most-used join key is not an entity

Three absences matter, in this order.

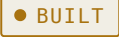
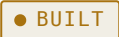
1 • No Metabolite. The effector of a dietary compound is frequently not the compound ingested. First-pass and gut-microbiome metabolism produce the actual actor — ellagitannins to urolithin A, daidzein to equol, glucosinolates to sulforaphane — and most polyphenols circulate as glucuronide or sulfate conjugates with materially different bioactivity. Every `Compound -ACTS_ON→ Target` edge therefore inherits a **parent-compound fallacy**, for exactly the compound class this graph is densest in.

2 • No Gene. Gene symbol is the join key behind `PATHWAY_INCLUDES_TARGET`, the 2.89M gene-mediated inference rows, and `targets.gene_symbol`. The graph reasons *through* genes constantly and cannot reason *about* them.

3 • No Nutrient. The mission sentence opens with "macronutrients," yet nutrition lives as a 90-key JSON blob on 647 Food nodes — unaddressable, unjoinable, invisible to traversal.

04 Edge schema — 21 declared, 6 confirmed live

RELATIONSHIP_TYPES · GROUPED BY BUILD PHASE					
EDGE	SRC → TGT	SOURCE TABLE	ROWS	STATE	CARRIED PROPERTIES
PHASE 0 — STRUCTURAL BACKBONE					
CONTAIN S_COMPO UND	Herb → Compound	herb_compound s	99,2 80	● LIVE	plant_part, conc. low/high ppm
FOUND_I N_FOOD	Compound → Food	compound_food s	4,14 9,54 1	● LIVE	content_value, unit, food_part
TARGETS _PROTEI N	Compound → Target	compound_targe ts	7,05 3	● LIVE	activity_value, activity_type
ASSOCIA TED_WIT H_DISEA SE	Target → Disease	target_disease s	795, 434	● LIVE	evidence_layer
TREATS_ SYMPTOM	Herb → Symptom	herb_symptoms	41,8 23	● LIVE	— none
PHASE 1 — ChEMBL BIOACTIVITY BRIDGE (ADR 0007)					
HAS_EVI DENCE	Compound → BioactivityEvid ence	bioactivity_ev idence	0	○ DORMANT	pchembl, activity_type
EVIDENC E_FOR_T ARGET	BioactivityEvid ence → Target	bioactivity_ev idence	0	○ DORMANT	assay_confidence, year
PHASE 2 — SYMPTOM BRIDGE					
MAPS_TO _DISEAS E	Symptom → Disease	symptom_diseas e_map	40 / 47	● BUILT	mesh, umls, icd10cm, match_score ≥ 0.5
PHASE 3 — EVIDENCE-TYPED DISEASE LAYER (ADR 0008)					
COMPOUN D_TREAT S_DISEA SE	Compound → Disease	compound_disea se_evidence	2.92 M †	● BUILT	pubmed_ids, source

EDGE	SRC → TGT	SOURCE TABLE	ROWS	STATE	CARRIED PROPERTIES
COMPOUND_MARKED_FOR_DISEASE	Compound → Disease	compound_disease_evidence	2.92M ↑	 BUILT	pubmed_ids
COMPOUND_INFERRED_DISEASE	Compound → Disease	compound_disease_evidence	2.92M ↑	 BUILT	inference gene, score, pubmed_ids
PHASE 4 – KEGG PATHWAY OVERLAY (ADR 0009)					
PATHWAY_INCLUDES_TARGET	Pathway → Target	kegg_pathway_genes	455	 BUILT	kegg_gene_id, gene_symbol
COMPOUND_IN_PATHWAY	Compound → Pathway	kegg_compound_pathways	0	 DORMANT	Blocked on empty compound_identity
CURATED SAFETY PANEL					
INTERACTS_WITH	Herb → Drug	hdi_safe_50.json	50	 LIVE	severity, mechanism_class, evidence_tier, citations
CONTRAINDICATIONS	Herb Compound → Condition Disease...	none	—	 DESIGNED	Polymorphic; no query defined
CLINICAL / TENANT LAYER – DECLARED, NO SOURCE					
INCLUDES	Protocol → Intervention	none	—	 DESIGNED	
USES	Intervention → Compound	none	—	 DESIGNED	
RESULTED_IN	Intervention → Outcome	none	—	 DESIGNED	
MEASURED_BY	Outcome → Biomarker	none	—	 DESIGNED	
INDICATES	Biomarker → Disease	none	—	 DESIGNED	

EDGE	SRC → TGT	SOURCE TABLE	ROWS	STATE	CARRIED PROPERTIES
SYNERGI ZES_WIT H	Compound → Compound	none	—	 DESIGNED	

† The three Phase-3 edges partition one 2,922,025-row table by `evidence_type`; the count is the table, not each edge.

05



Read the dashed edges as the honest picture: everything built after 2026-05-07 is unconfirmed in the deployed graph, and three edge types return zero rows by construction. The single richest edge in the entire schema — the only one carrying `mechanism_class`, `severity` and `evidence_tier` — is the 50-row hand-curated herb-drug panel.

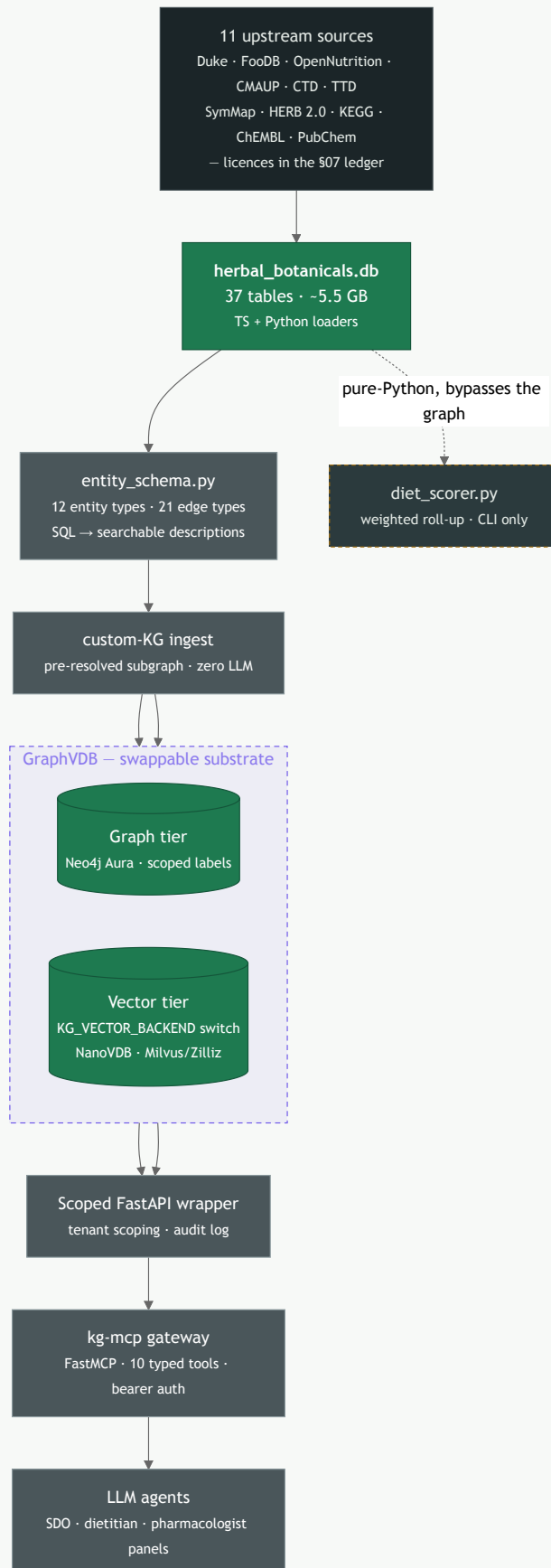
06 Data construction

The construction philosophy is **structured-first**: normalize every source into one SQLite database, then push typed entities and edges into the GraphVDB through its custom-KG ingest call, which bypasses LLM extraction entirely — 100% fidelity, zero token cost. Semantic search is a *vector index layered over* those deterministic edges, not a producer of them.

That ingest call is the one place the substrate leaks into our code. It is a single verb — *write this exact set of nodes and edges, do not infer any* — and any graph-plus-vector store that can accept a pre-resolved subgraph satisfies it. Keeping the contract that narrow is what makes the vendor swappable; the current binding implements it as

```
ainsert_custom_kg().
```

INGEST PIPELINE – SOURCES TO QUERYABLE GRAPH



Note the dashed branch. `diet_scorer.py` — the read-side capstone that turns `(food, grams)` into ranked target/disease/pathway predictions — reads **SQLite directly and never touches the graph**. It was kept off the MCP surface deliberately, to respect a `FORBIDDEN_USECASE_VERBS` thin-adapter constraint. The consequence is that the system's most mechanistically interesting computation is invisible to every agent that talks to the gateway, and its five evidence weights live in a Python dict rather than on the edges they weight.

WHERE THE SEMANTICS LEAK OUT

Two losses happen at the SQLite → graph boundary, both in the same direction:

- `compound_targets` has an `interaction_type` column. The `TARGETS_PROTEIN` query selects `activity_value` and `activity_type` and **drops it**. The one direction-bearing field in the binding table never reaches the graph.
- `bioactivity_evidence` captures `pchembl`, `value`, `units`, `relation`, `assay_confidence` — a complete potency record with **no action field at all**. ChEMBL knows how hard something binds; this schema never asks what happens next.

07 Source ledger

INGESTED SOURCES · LICENCE POSTURE · LAST REFRESH				
SOURCE	LICENCE	CONTRIBUTES	LAST REFRESH	COMMERCIAL?
Dr. Duke's Phytochemical DB	CC0	Herb, Compound, Food, Symptom backbone	2026-04-12	
FooDB	CC-BY-4.0	4.13M compound–food edges	2026-04-12	
OpenNutrition (USDA FDC core)	CC-BY-4.0	90-key nutrition on 647 Foods	2026-04-29	
CMAUP v2.0	Academic	Targets, compound–target, plant–disease	2026-04-29	
SymMap 2.0	Academic	TCM symptoms, bilingual crosswalk, genes	2026-04-26	
HERB 2.0	Academic	1.8M herb–disease evidence rows	2026-04-12	
TTD	Academic	Literature refs only; bulk table refused as off-mission	—	
CTD	Public files	2.92M compound–disease evidence w/ PubMed	2026-05-08	
KEGG	Academic only	370 pathways, 39,340 pathway–gene links	2026-05-08	
HDI-Safe-50 (curated)	Derived, cited	50 herb–drug edges — the only mechanism-bearing edges	2026-05-01	

PROVENANCE DRIFT

`DATASET_PROVENANCE.md` is stamped **2026-05-01** and documents per-source refresh cadences that have not been exercised since. For a graph whose entire pitch is evidence-grading, unexercised pins are a credibility gap that a reviewer will find faster than we will. It is cheap to close and should ship with the rename.

08 Potency without polarity

Here is the whole argument in one line: **the graph knows how strongly a compound binds and never what the binding does.**

Every quantitative edge in the schema is a magnitude with no sign. `TARGETS_PROTEIN` carries `activity_value`. `BioactivityEvidence` carries `pchembl`, `value`, `units`. None of them record whether the compound *inhibits*, *activates*, *antagonises*, or merely *occupies* the target. The one place mechanism appears anywhere in the system is `mechanism_class` on the 50 hand-curated herb-drug edges — 50 out of roughly 5 million.

That absence propagates directly into the read side. `diet_scorer.py` aggregates exposure across compounds and rolls it up per target with strictly non-negative weights and no negative-evidence subtraction — its own ADR lists this as an invariant. Two compounds in the same meal, one inhibiting a target and one activating it, sum to a *stronger* predicted effect. There is no representation in which they could cancel, because polarity was never ingested.

And it plausibly reaches the paper's headline finding. Paper 1 established that grounded panels cite non-existent chains in 40% of predictions, with 13 of 40 citing chains when retrieval returned nothing. A retrieval layer that returns unsigned, unranked, mechanism-free edges gives a model very little to be faithful to. Fabrication and an under-specified mechanism layer are not the same bug, but they are adjacent, and no experiment so far has separated them.

POLARITY ALONE IS NOT ENOUGH — THE EXPOSURE REGIME

Adding action verbs from a drug-pharmacology vocabulary solves the sign and introduces a subtler failure. That vocabulary is validated at **pharmacological doses**. Dietary bioactives operate in a different epistemic regime: most bind weakly and promiscuously, and post-prandial systemic concentrations are typically nanomolar to low micromolar — frequently orders of magnitude below the IC50 and Ki values ChEMBL records.

The consequence is sharp. A graph that faithfully records "*quercetin inhibits PI3K, IC50 = X μ M*" and lets an agent traverse that edge will emit chains that are mechanistically correct-looking and physiologically implausible at dietary exposure. Relative to today's unsigned graph, that is **a more legible hallucination substrate** — better-formed claims, no better grounded.

So exposure plausibility is not a downstream generalization of the MoA layer. **It is the validity gate on it.** A mechanism node without a concentration-context qualifier answers "what does binding do in an assay," not "what does this food do in a person" — and the second question is the one in the title.

SOURCING CAVEAT — STATED PLAINLY

Achievable-Cmax data for dietary compounds has **no structured source among the ten currently ingested**. It sits in primary pharmacokinetics literature with no ChEMBL-equivalent, so a true Cmax join is a data-acquisition project, not a schema change. The buildable v1 is a coarse ordinal — assay context (cell-free / cell-based / in vivo) plus an assay-concentration bucket — used as a plausibility proxy. [A3](#) specifies the qualifier; the Cmax join is named as its own follow-on rather than assumed away.

THE PARENT-COMPOUND FALLACY

Equally load-bearing: for a diet-origin graph the effector is often a metabolite, not the ingested compound (§03). Modelling `Compound -ACTS_ON→ Target` with no metabolism hop is systematically wrong for the polyphenol class the graph is densest in. [A7](#) adds Metabolite and `METABOLIZED_T0` with a `context: hepatic | microbial` qualifier — carrying the same caveat, since none of the ten current sources holds those edges either.

THE REFRAME

Semantic context as the driver, not the veneer

Today the pipeline is structured-first with a semantic index bolted on top: SQLite rows become templated description strings, which become embeddings. Natural language is an *output* of the structured layer — which is precisely why it cannot be wrong in ways the rows are not.

The proposal inverts the dependency. A **mechanism-of-action layer** — action semantics, pharmacological class, mechanistic narrative — becomes a first-class, addressable part of the graph, and the quantitative layers (potency, exposure, citation count) attach to it as measurements *of* a stated mechanism. Natural language stops being a rendering of the rows and becomes the interpretable spine the rows are evidence for.

Mechanism text is sourced, never generated

Narrative and mechanism-text fields come from curated upstream text carried verbatim with its citation, or they do not exist. Any narrative synthesised at query time must be reconstructible from cited edges, and an agent must be able to name the edges it came from.

Without this clause the reframe reintroduces exactly the fabrication surface Paper 1 measured — relocated from retrieval time into the knowledge layer, where it is harder to detect. A project whose headline finding is citation fabrication cannot ship an LLM-authored mechanism corpus.

THE THESIS IS TESTABLE

If the mechanism-gap hypothesis is right, adding signed, tiered, cited, exposure-qualified edges should **measurably reduce the 40% non-existent-chain citation rate**. That is a clean A/B: same panels, same questions, retrieval with and without the new layers. [A10](#) makes it the acceptance gate rather than a hoped-for side effect.

09 Proposed amendments

Ordered by dependency, not by ambition. A0 through A2 are prerequisites — nothing downstream is trustworthy until they land. A3 through A6 are the MoA layer proper. A7 through A9 follow from the rename, and are cheap only if they ride the same sweep.

A0 · Prerequisite

● PROPOSED

Re-snapshot the live graph — then make drift impossible

Run `graphvdb/generate_snapshot.py` and diff against the 2026-04-29 baseline. This is minutes of work and it determines whether every amendment below is an additive migration or sits behind a full re-ingest. Requires Aura credentials from the Infisical project this identity cannot currently read — an access grant is the actual blocker.

Then make it a gate, not an event. The three-tier drift happened because snapshotting is manual. The durable fix is a CI job that diffs snapshot output against a committed baseline and fails the build on drift. Without it, A0 gets re-proposed in six months under a different name.

WHY FIRST

Eleven of twenty-one edge types have an unknown deployment state. Designing on top of that is designing on a guess.

A1

● PROPOSED

Activate `compound_identity`

Run the Phase 1 PubChem/UniChem/ChEMBL bridge end-to-end. It has been built and bug-fixed twice but never executed to completion, so `compound_identity` is an empty schema. Three edge types are dormant behind it and the food $\cap$ target intersection is stuck at 565 compounds against a $\geq 1,500$ target.

UNLOCKS

`HAS_EVIDENCE`, `EVIDENCE_FOR_TARGET`, `COMPOUND_IN_PATHWAY`; the ChEMBL potency corpus; ADR 0007's own $\geq 70\%$ cross-ref acceptance gate, still unmeasured.

A2

● PROPOSED

Stop dropping `interaction_type`

A one-line change to the `TARGETS_PROTEIN` query. The column already exists in `compound_targets` and is discarded at ingest. Whatever CMAUP's coverage turns out to be, it is the cheapest polarity signal available and it establishes the property name the richer MoA layer will extend.

COST

One SQL line plus a re-ingest of one edge type. Measure the fill rate before designing around it.

Reify mechanism of action — with context qualifiers from day one

Introduce MechanismOfAction as an addressable node that Compound and Target attach to, rather than binding to each other directly. Reification is what lets one compound–target pair hold several competing mechanism claims from different sources without one silently overwriting another.

REQUIRED FIELD	WHY IT CANNOT WAIT
<code>action</code>	Polarity and modality — adopt, don't invent (below)
<code>pharmacological_class</code>	Groups mechanisms above the single-target level
<code>species</code> / <code>system</code>	CTD and ChEMBL mix human, rodent and cell-free. A rodent in-vivo agonism claim served to a human question is a silent category error.
<code>assay_context</code>	Cell-free / cell-based / in vivo — also the cheapest exposure-plausibility proxy
<code>exposure_plausibility</code>	Ordinal tier: is the assay concentration within reach of dietary intake?
<code>confidence_tier</code>	Curation strength, distinct from citation count
<code>narrative</code> + <code>citation</code>	Sourced verbatim, never generated (§08)

Three new edges: `Compound` –EXERTS→ `MechanismOfAction`, `MechanismOfAction` –ACTS_ON→ `Target`, `MechanismOfAction` –EVIDENCED_BY→ `BioactivityEvidence`. The existing `TARGETS_PROTEIN` edge stays as the fast unsigned path so nothing downstream breaks.

Adopt the action vocabulary, do not invent one. ChEMBL mechanism `action_type` and the IUPHAR/BPS Guide to PHARMACOLOGY annotations already exist, are permissively licensed, and give external interoperability. Inventing a closed verb list that later needs mapping to them is avoidable debt — and would contradict §12's own advice to prefer open substrates.

Reify interaction claims the same way. `SYNERGIZES_WITH` as a bare

`Compound`→`Compound` edge has exactly the defect diagnosed for `TARGETS_PROTEIN`: synergy is a claim about a pair *in a context* — dose ratio, model, endpoint. If the interaction layer is built (Q1), reify it rather than reviving the flat dormant edge.

WHY AN ENTITY AND NOT AN EDGE PROPERTY

Mechanism claims disagree across sources, carry their own provenance, qualifiers and confidence, and need to be cited individually. A property on an edge cannot hold two conflicting claims; a node

A4

● PROPOSED · REORDERED IN V2

Open substrate first; proprietary as licence-gated enrichment

Populate the MoA layer from ChEMBL MoA tables and IUPHAR/BPS first — thinner than a proprietary source, clear-licence, and already the vocabulary authority under A3. Treat a DrugBank-class source as enrichment *on top of* that spine, following the KEGG precedent exactly: one build-time entry point, one toggle, an explicit provenance entry, a documented degradation path when the licence is absent.

WHY THIS ORDER

A clear-licence spine with proprietary enrichment degrades gracefully; the reverse does not. Six of ten current sources are already academic-only.

A5

● REWRITTEN IN V2

Weight at query time; never stamp policy onto data

The five weights in `diet_scorer.py` (direct binding 1.00, therapeutic 0.90, marker 0.70, gene-inferred 0.50, pathway 0.60) encode a real epistemology and are invisible to retrieval. **v1 of this document proposed stamping them as an edge property at ingest. That was wrong** — it conflates provenance with policy, makes every weight revision a re-ingest, and stops two tenants with different risk postures from disagreeing.

The correct split: A6 stamps raw attributes; the scorer reads attributes and applies weights **at query time**, with the weight set exposed as a versioned parameter and overridable per tenant. The scoped wrapper already enforces tenant scopes, so the mechanism exists.

ANSWERS Q4

Yes, the scorer should move onto the graph — as a traversal-plus-weighting function over well-attributed edges, not as baked-in numbers.

A6

● PROPOSED

Uniform provenance stamp and build gate

Today `pubmed_ids` exists on the Phase-3 evidence edges (94% fill) and nowhere else; `TREATS_SYMPTOM` carries no properties whatsoever. Require `source`, `evidence_tier`, `citation`, plus `assay_type` and `species` where applicable, on every ingested edge — with a build gate that fails on unstamped ones. Add `licence_tier` alongside so a tenant scope can filter to commercially-clear edges.

Same mechanism, second check: **refresh-cadence compliance** as a build gate, so the stale pins in §07 fail loudly instead of sitting in a doc.

PRECONDITION FOR

A cite-or-abstain retrieval contract, and for A5's query-time weighting — which needs attributes to weight over.

A7

● PROPOSED · REORDERED IN V2

Add Metabolite; declare Gene and Drug; promote Nutrient

Metabolite + **METABOLIZED_TO** with `context: hepatic | microbial`. Without it the MoA layer inherits the parent-compound fallacy from the occurrence layer. Schema is the easy half — none of the ten current sources carries these edges, so this implies an ingest project and is sequenced accordingly.

Gene and Drug already exist as live undeclared nodes — regularise them.

Nutrient comes out of the

`nutrition_100g` blob.

RENAME-DRIVEN

Generalising past diet makes Gene the natural mechanistic hub and Drug a peer exposure route, not an exception.

A8

• SCOPE REDUCED IN V2

Generalise Food to ExposureRoute

Food is one way a compound enters a body — alongside supplement, standardised extract, and drug. Keep Food as a subtype; introduce the parent so dose and matrix attach to the route rather than being hard-coded to the dietary case.

CHANGED

The plausibility *qualifier* that gates mechanism claims moved into A3, where it belongs. A8 is the carrier that makes it computable per route — not the place the gate lives.

A9 · Naming

• PROPOSED

Name the retrieval substrate by role — `lightrag/` → `graphvdb/`

A vendor name currently sits in a structural path, twice over: the vendored upstream submodule and our own module directory are both called `lightrag/`, so *which one* is a question every reader of this tree has to answer. Rename our module to `graphvdb/`, leave the vendored submodule under its real upstream name, and let the directory boundary state which code is ours and which is theirs.

The coupling to abstract is thin and already understood. Ingest is one call — write this pre-resolved subgraph, infer nothing. Read is the scoped wrapper plus the five query modes. Storage is already vendor-neutral on one tier: `KG_VECTOR_BACKEND` switches NanoVDB for Milvus/Zilliz, which is exactly this move applied to the vector side. A9 finishes it on the graph side and gives the abstraction a name the docs, the ADRs and the MCP tool descriptions can all share.

WHY BUNDLE IT WITH THE RENAME

It is the same sweep, the same reviewers, and the same regression surface. Splitting it means touching `lightrag/` imports twice. Caveat: a role name only earns its keep if the boundary is enforced — if substrate-specific calls keep leaking into ingest and query code, `graphvdb/` is a rename that buys nothing. Pair it with a thin interface module and a test that fails on direct vendor imports outside it.

Pre-register the fabrication-rate eval as the acceptance gate

Paper 1 measured a 40% non-existent-chain citation rate and a citation-faithfulness score of 0.66. **That metric is the natural success criterion for this entire program.** Pre-register it: same panels, same questions, retrieval with and without the A2/A3/A6 layers, with the target and the analysis fixed before the run.

Without it, the risk is shipping a beautiful mechanism layer whose effect on agent faithfulness is never measured — the same unexercised-pins credibility gap this document flags for provenance in §07.

COSTS, NAMED

Requires freezing the 40×7 gpt-oss-120b matrix as a regression baseline, and consumes real inference budget per run. It is also only interpretable after A0 and A2: if the two open gateway defects — degraded NL retrieval, and the 400 on neighbourhood queries — are still live, part of the 13/40 "cited when retrieval returned nothing" is explained with no MoA hypothesis at all.

Ingest negative and inactive results

Two one-sidedness problems compound each other. **Citation count imports publication bias** — `pubmed_ids` at 94% fill is good provenance, but count-of-citations as a weight input rewards positive results, which are cited more, and the diet scorer's `citation_factor` uses exactly this. **Negative and inactive assays are absent from the schema entirely** — ChEMBL carries inactive rows.

WHY IT LANDS WITH A3

Ingest them, or the polarity layer is itself one-sided: it will record that compounds act, and never that they were tested and did not.

PROGRAM ORDER

Dependency-ordered, revised in v2 to put provenance before mechanism and exposure inside it.

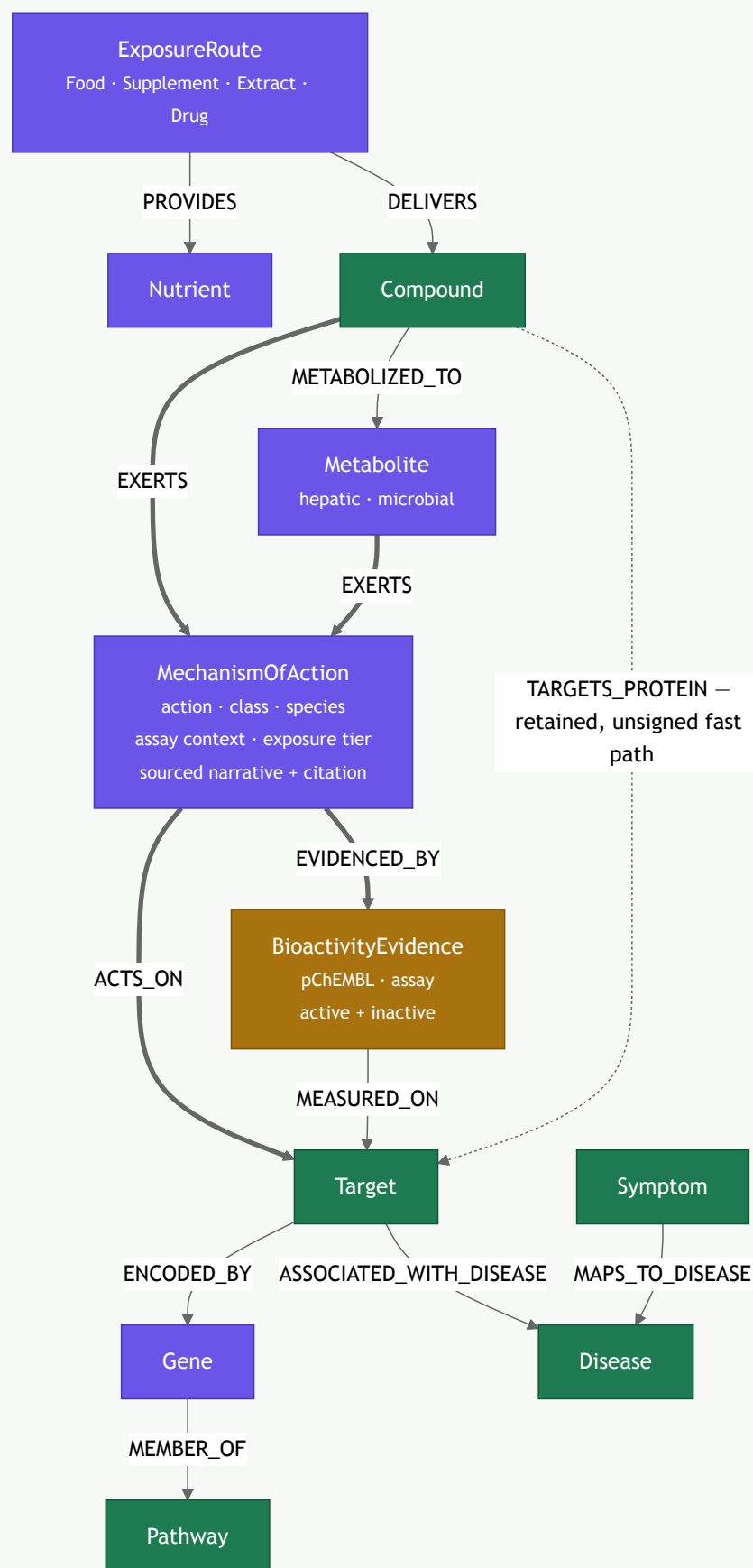
SEQUENCED PROGRAM

#	AMENDMENT	WHY HERE
1	A0 re-snapshot + drift gate	Everything downstream is designed on a guess until this lands

#	AMENDMENT	WHY HERE
2	A1 activate <code>compound_identity</code>	Unblocks three dormant edge types and the ChEMBL corpus
3	A2 stop dropping <code>interaction_type</code>	One line; cheapest polarity signal; sets the property name
4	A6 provenance stamp + build gate	Mechanism claims are worthless unattributed — attribution precedes them
5	A3 (+A8) MoA with exposure and context qualifiers	The core; qualifiers are far cheaper now than as a retrofit
6	A11 negative / inactive results	Lands with A3 or the polarity layer ships one-sided
7	A4 open substrate, then licensed enrichment	Vocabulary authority is an A3 input; the licensed layer is additive
8	A5 query-time weighting	Needs A6's attributes to weight over
9	A7 entity regularisation + Metabolite	Metabolite implies its own ingest project; sequence accordingly
10	A9 naming	Rides the repo-rename sweep
—	A10 pre-registered eval	Registered <i>before</i> step 5; measured after

10 Target schema

PROPOSED SHAPE – MOA REIFIED BETWEEN COMPOUND AND TARGET



Two load-bearing changes. The double arrows replace a single unsigned edge with a **claim** — action, species, assay context, exposure tier, sourced narrative, its own citations — that can coexist with a contradicting claim instead of overwriting it. And Metabolite also **EXERTS**, so the effector of a dietary compound can be modelled as the thing that actually acts. Everything else is regularisation: naming what the graph already does implicitly.

11 Qualitative $\rightleftharpoons$ quantitative

The brief asks for a graph that is both. Worth being precise about which layer supplies which, and where each is currently thin.

QUALITATIVE – THE INTERPRETABLE SPINE

- Action semantics, adopted vocabulary
● PROPOSED
- Pharmacological class, sourced narrative
● PROPOSED
- Species / assay context / exposure tier
● PROPOSED
- Metabolism context — hepatic, microbial
● PROPOSED
- `mechanism_class` on herb–drug edges
● LIVE · 50 EDGES
- Entity descriptions rendered for embedding
● LIVE
- Ontology anchors — MeSH, UMLS, ICD-10, HPO
● BUILT
- Bilingual EN/CN/Pinyin herb crosswalk
● LIVE

QUANTITATIVE – THE MEASUREMENT LAYER

- pChEMBL, IC50/Ki, assay confidence
○ DORMANT
- Binding `activity_value`
● LIVE · 7,053
- Compound content per food, mg/100 g
● LIVE · 4.15M
- Gene-mediated inference score
● BUILT · 2.89M
- PubMed citation counts
● BUILT · PUBLICATION-BIASED
- Exposure roll-up from grams consumed
● BUILT · CLI ONLY
- Negative / inactive assay results
○ ABSENT
- Cmax, bioavailability, dose–response
○ ABSENT · NO SOURCE

The asymmetry is the point. The quantitative column is populated or one ingest away from populated; the qualitative column is 50 curated edges and a set of templated description strings. A mechanism-of-action knowledge base is mostly the left column, and the left column is where almost nothing has been built.

Two entries in the right column are marked absent for different reasons, and the difference matters for planning. Negative and inactive results exist upstream in ChEMBL and simply were not ingested — that is a decision to reverse (A11). Achievable Cmax has *no structured source at all* among the ten in §07, so it is an acquisition problem wearing the costume of a schema gap.

12 IP posture for a proprietary MoA layer

Six of the ten ingested sources are academic-use-only, and KEGG is explicitly commercial-licence-required. Adding a DrugBank-class layer makes the licence question structural rather than incidental, so the graph should be able to state its own posture rather than have it live in a doc.

- **Keep the KEGG precedent.** One build target per restricted source, one toggle, a documented degradation path. A commercial deployment builds without them and loses coverage rather than shipping a licence breach.
- **Stamp licence tier on the edge.** If `source` and `evidence_tier` become mandatory under A6, add `licence_tier` alongside. A tenant scope can then filter to commercially-clear edges at query time — the scoped wrapper already enforces per-tenant scopes, so the mechanism exists.
- **Separate ingest from redistribution.** Deriving mechanism claims and serving them through an MCP gateway is a different licence question from mirroring the source. Worth counsel review before the first proprietary overlay lands, not after.
- **Prefer an open substrate where one exists.** ChEMBL mechanism-of-action tables and Open Targets both carry action-type data under permissive terms. They are thinner than DrugBank, but a clear-licence spine with a proprietary enrichment on top degrades far more gracefully than the reverse.

13 Open questions

Q1 "Interactivity" — which one?

It reads two ways and they build differently. *Interaction knowledge* — herb–drug, compound–compound synergy, contraindication — extends the 50-edge HDI panel and the dormant `SYNERGIZES_WITH` / `CONTRAINDICATES` types into a real layer. *Interactive querying* — conversational multi-turn traversal — is a gateway and agent concern, not a schema one. This document assumes the first; say if it is the second, or both.

Q2 Does the rename widen ingest scope now, or later?

Generalising the mission reopens the bulk TTD drug–disease table and the deferred TCMSP / STITCH / DisGeNET / BATMAN-TCM tier. Ingesting them is a large amount of work; refusing them under a name that no longer says "diet" is incoherent. A dated ADR either way is enough.

Q3 How much proprietary MoA, and when?

A4 now says open substrate first — ChEMBL MoA and IUPHAR/BPS as the spine, a DrugBank-class source as enrichment. Confirm that framing before any procurement conversation, because it changes what the licensed layer has to deliver to be worth buying.

Q4 Should the scorer move onto the graph?

A5 answers yes — as query-time weighting over attributed edges. It still deserves its own ADR, because it reverses the thin-adaptor constraint that kept `diet_scorer.py` off the MCP surface in the first place.

Q5 Does the mechanism gap explain part of the fabrication finding?

Paper 1 reports 40% of grounded predictions citing non-existent chains, with 13 of 40 citing when retrieval returned zero. Two open gateway defects from May — degraded NL retrieval and a 400 on neighbourhood queries — have never been verified as fixed. A0 and A2 are cheap and separate "the model fabricates" from "retrieval returned nothing useful to be faithful to." A10 then makes the answer measurable rather than argued.

Q6 Is an exposure-plausibility tier worth building before a real Cmax source exists?

§08 argues yes, because retrofitting the field is a re-ingest and the coarse assay-context proxy is free once A3 lands. The counter-argument deserves a hearing: a coarse tier can produce false confidence, and a wrong plausibility label may be worse than an absent one.

Q7

Who owns the fabrication-rate baseline?

A10 requires freezing the 40×7 matrix as a regression artifact with a standing budget for re-runs. That is an ongoing cost with an owner, not a one-off task. If nobody owns it, A10 becomes another unexercised pin.

14 Review response

External design review received 2026-08-01. This section records what changed and, where I qualified rather than simply accepted, why.

ACCEPTED IN FULL

V1 → V2 CHANGES	
POINT	CHANGE
A5 was wrong	Stamping scoring weights at ingest conflates provenance with policy, makes every revision a re-ingest, and blocks tenants from disagreeing. Rewritten as query-time weighting over A6 attributes. v1's A5 and A6 partly contradicted each other; A6's discipline wins.
Exposure is a validity gate	Promoted from A8 into A3's required fields. The "more legible hallucination substrate" framing is the sharpest point in the review and is now §08.
Metabolite outranks Nutrient	The parent-compound fallacy is the load-bearing absence for a diet-origin graph. Now A7's headline, with <code>METABOLIZED_TO</code> and a context qualifier.
Adopt the vocabulary	Inventing a seven-verb list contradicted this document's own §12. ChEMBL <code>action_type</code> / IUPHAR-BPS are now A3's authority.
Species + assay context day one	Both added to A3 — retrofitting either is a re-ingest.
Reify <code>SYNERGIZES_WITH</code>	Same defect as <code>TARGETS_PROTEIN</code> , same fix. Noted in A3 and Q1.
Narrative provenance	§08 now carries a non-negotiable: mechanism text is sourced, never generated. v1's phrasing could be read either way, and one reading contradicts Paper 1's central finding.
A0 is a gate, not an event	CI drift check added.
Publication bias + missing inactives	New A11.
Cadence as a build check	Folded into A6 — one mechanism, two checks.

ACCEPTED WITH A QUALIFICATION

Exposure plausibility. The gate is right; the data is not there. Achievable-Cmax for dietary compounds has no structured source among the ten ingested, so v1 of the field must be the coarse assay-context proxy the review itself suggests, with the Cmax join named as a separate acquisition project. Saying this matters — otherwise the gate reads as a schema change when it is substantially a sourcing problem, and it gets estimated as the former.

Metabolite. Same shape. The schema is the easy half; microbial and phase-II metabolism edges are in none of the current sources, so A7 implies an ingest project and is sequenced late in the program order for that reason, not because it matters less.

The eval gate. Adopted, with costs named: freezing the 40×7 gpt-oss-120b matrix as a regression baseline, real inference budget per run, and interpretability that depends on A0 and A2 landing first to remove the retrieval-defect confounder.

No disagreement recorded on the rename analysis, on ADR 0011 first, or on the core diagnosis.

How this was grounded. Entity and relationship counts read from `graphvdb/entity_schema.py` at commit `db3c740`. Row counts from `docs/KG_COMPLETENESS_AUDIT.md` Phase 2–5 closeouts and ADRs 0007–0010. Live-graph state from `research-journal/shared/scope-state-snapshot.md`, generated 2026-04-29 – the most recent snapshot in the repo. Source licences and refresh dates from `docs/DATASET_PROVENANCE.md`, stamped 2026-05-01. Name-debt counts from `git grep -l` over tracked files. Gateway health verified live: `GET /health` → 200.

On naming. `GraphVDB` is this document's role name for the graph-plus-vector substrate and is not yet a name in the codebase – see the nomenclature note above and amendment A9. Every `graphvdb/...` path cited here resolves to `lightrag/...` in the tree at `db3c740`; the counts and line numbers are read from those files unchanged.

What is not verified. The current contents of the Neo4j Aura instance. Credentials live in an Infisical project this identity is not a member of, so every "Live" state in this document is as-of the 2026-04-29 snapshot and every post-Phase-2 edge is marked Built rather than Live for that reason. Amendment A0 exists to close this gap first.